



SORO ONANSONGUI ELIXIR

ACTUARY IN TRAINING

+225 07-59-987-589 · +225 07-07-336-135
elixirsoro1@icloud.com
Cocody, Abidjan, Côte d'Ivoire

PROFILE

With a dual academic background in economics and actuarial science, I combine mathematical rigor with advanced Data Science skills. Passionate about risk modeling and data analysis, I am proficient in programming tools (R, Python) to solve complex problems in ALM, pricing, and Machine Learning. I am seeking an opportunity to apply these skills to risk management and quantitative analysis.

PROFESSIONAL EXPERIENCE

Actuarial & AI Automation Intern

AMGS (African Mutual Global Services) · Since July 2026 · Reinsurance Department (CIMA/OHADA zone)

ELIXIR Pipeline — Facultative Reinsurance Automation

- designed and led an 11-agent orchestrated AI system automating the end-to-end facultative reinsurance quotation cycle.
- Reliability** : zero data fabrication, zero output sent without human validation — two safeguards made verifiable through computation.
- Governance** : SHA-256 fingerprint governance preventing agent self-modification; integrated 124 ceding companies and 600 reinsurers/captives.
- Test suite** : 19 automated tests and 12 functional cases; final run: 9 files produced, 100% of deterministic checks passed.

Automated Medical Invoice Control Agent

AMGS SANTÉ

- designed and led a software agent automating the receipt, OCR extraction, and reconciliation of medical invoices with Odoo.
- Safeguards** : mandatory human validation, audit logging, idempotency.
- Development** : specifications, code review, and fixes directed through generative AI tools (Claude, Claude Code).
- IT coordination** : Odoo XML-RPC integration, Microsoft Graph/Entra ID migration; UAT via validation dashboard and 150+ automated tests.

ACADEMIC PROJECTS

Design of an Economic Scenario Generator (ESG) and ALM Model

2025

Master's Project · R, Shiny

- built an R Shiny application for Asset-Liability Management (ALM) in Life Insurance.
- Stochastic modeling** : implementation of the Vasicek model for interest rates and Geometric Brownian Motion (GBM) for the CAC40.
- Risk management** : simulation of economic shocks (e.g. a COVID-19-type crisis) with impact analysis on the asset-liability balance.
- Compliance** : alignment with Solvency II requirements (SCR calculation).

Predictive Modeling & Machine Learning — Vehicle Pricing

2025

Data Science Project · Python

- built complete Machine Learning pipelines with Scikit-learn on the car_price_prediction.csv dataset.
- Approaches** : regression (Lasso, Ridge, ElasticNet) for price prediction and classification for range categories.
- Data Engineering** : missing-value treatment (KNN Imputer), categorical encoding (OneHotEncoder), and feature engineering.
- Results** : Accuracy: 0.8099 (81.0%) · F1-Score: 0.8071 (80.7%).

Interactive Dashboard for Insurance Portfolio Monitoring

2025

R Shiny Project · R, Plotly

- built an interactive dashboard for technical portfolio monitoring.
- Key indicators** : automated calculation and visualization of Loss/Premium ratios (S/P) by year and by policyholder.
- Visualization** : dynamic, animated charts using ggplot2, plotly, and shinydashboard.

REFERENCES

Anouar Meynaoui

Senior Lecturer, Université de Rennes 2
anouar.meynaoui@univ-rennes2.fr

Xavier Gruchet

Actuary, Lecturer, Université de Bretagne Occidentale
xgruchet@gmail.com

Komla Tekpo Dzidzinyo

Data Scientist, SANLAM Côte d'Ivoire
kodzidzinyo2089@gmail.com

Promethée Kouame

Actuary
kpromethee@yahoo.fr

EDUCATION

Master's (M2) — Mathematics and Applications, Actuarial Science

Université Félix Houphouët-Boigny · 2025–2026 · Cocody, Abidjan

Master's (M1) — Mathematics and Applications, Actuarial Science

Université Félix Houphouët-Boigny · 2024–2025 · Cocody, Abidjan

Bachelor's — Mathematics, Actuarial Science

Université Félix Houphouët-Boigny · 2021–2024 · Cocody, Abidjan

Bachelor's — Economics, Development Economics

Université Félix Houphouët-Boigny · 2018–2021 · Cocody, Abidjan

Baccalauréat, Series D (Science)

Lycée Moderne Yopougon-Andokoi · 2016–2017 · Yopougon, Abidjan

SKILLS

Python (Pandas, Scikit-learn, Numpy)
R (Shiny, Tidyverse)
Excel (VBA), LaTeX
Time Series
Data Analysis — Data Science
SQL

LANGUAGES

French — Native
English — Technical proficiency